昇思 MindSpore

MindSpore Transformers

MindSpore Transformers 大模型系列课程

MindSpore Transformers LLM推理迁移与实践

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MindSpore Transformers 大模型系列课程

专题一：套件介绍

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《MindSpore Transformers环境安装与镜像制作》

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1. MindSpore Transformers Mcore推理架构介绍

MindSpore Transformers大模型推理技术架构全新升级

离线推理 & 在线部署

- Hugging Face权重加载
- 配置化模型开发
- 注册机制一键部署

Models

Hugging Face模型配置

Hugging Face
配置原生加载：0 Day迁移

TransformerConfig
Layer API接口
GPTModel

Megaton-LM API对齐：Cell级精度比对

- DeepSeek-V3
- Qwen3/Qwen3-MoE
- Llama3.1/3/2
- GLM4/4.5
- ...

模块化模型接口（Module Spec模块化组装）

- GPTModel
- TransformerBlock
- TransformerLayer
- MultiTokenPrediction
- MoELayer
- MLP
- MultiLatentAttention
- FlashAttention
- ColumnParallelLinear
- RowParallelLinear
- VocabParallelEmbedding
- ...

多维混合并行

- 模型并行(TP)
- 数据并行(DP)
- 专家并行(EP)
- 流水线并行(PP)
- 序列并行(SP/CP)

服务化部署 | 榜单评测

vLLM-MindSpore Plugin/ SGLang-MindSpore Plugin

- 接口兼容
- 组件解耦
- 最小化侵入式修改

AISBench基准工具/ OpenCompass榜单评测

- 支持20+主流数据集
- 服务化数据集精度验证
- 延迟和吞吐性能压测

模型快速迁移 | 精度比对

- Hugging Face配置原生加载
- MCore接口精度对齐vLLM
- Hugging Face Tokenizer无缝对接
- 快速权重转换
- Hugging Face权重在线加载
- Cell级dump工具

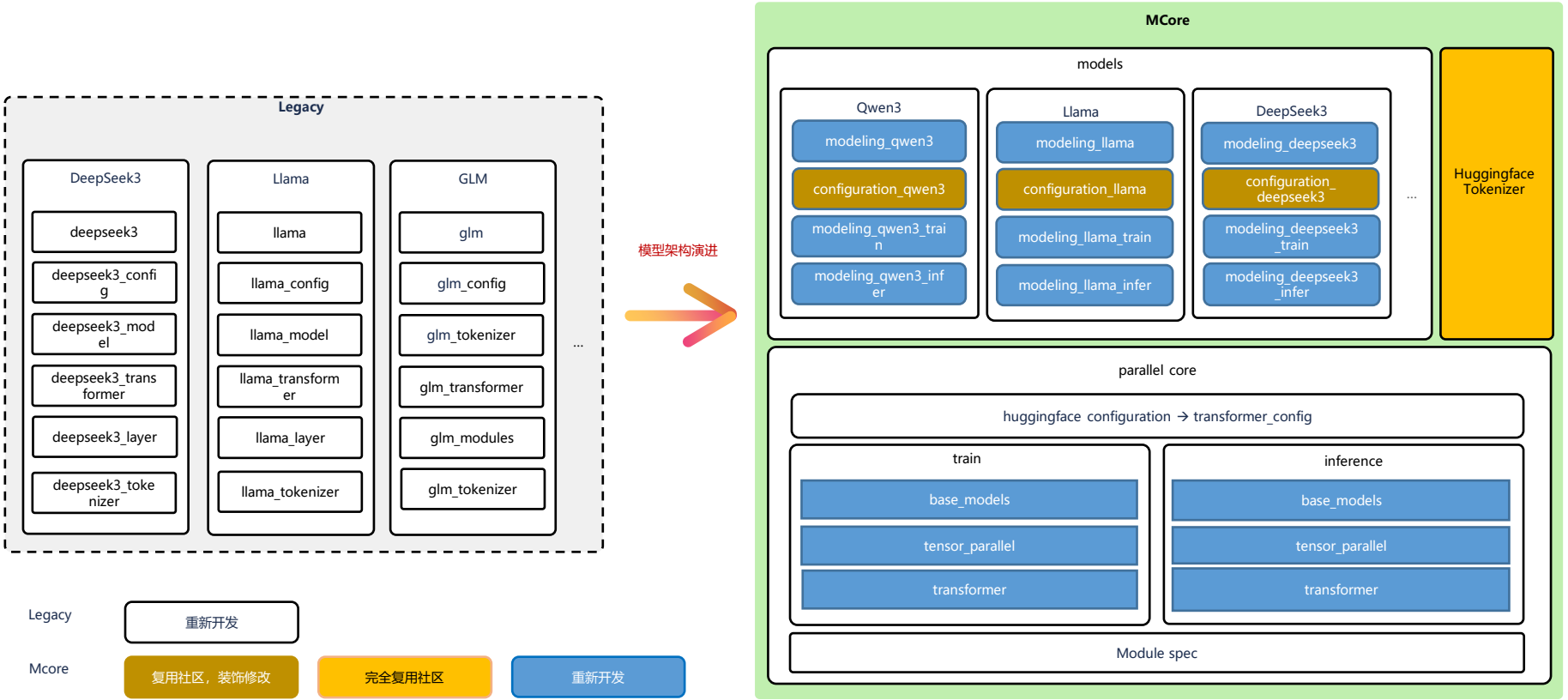
量化推理 | 高性能算子

- 支持金箍棒量化校准
- 支持HF/ModelSlim量化权重推理
- A16W8/A8W8/A8W4/C8...

- 对接昇腾ATB和ACLNN算子库
- 自研融合算子优化
- 自定义算子接入（能力构建中）

模型开箱即用 | **Hugging Face主流模型0day迁移** | **高精度/性能推理**

MindSpore Transformers Mcore模型库组件——整体架构



MindSpore Transformers Mcore推理模型库 ——整体架构

模型层 (parallel_core/inference/base_models)

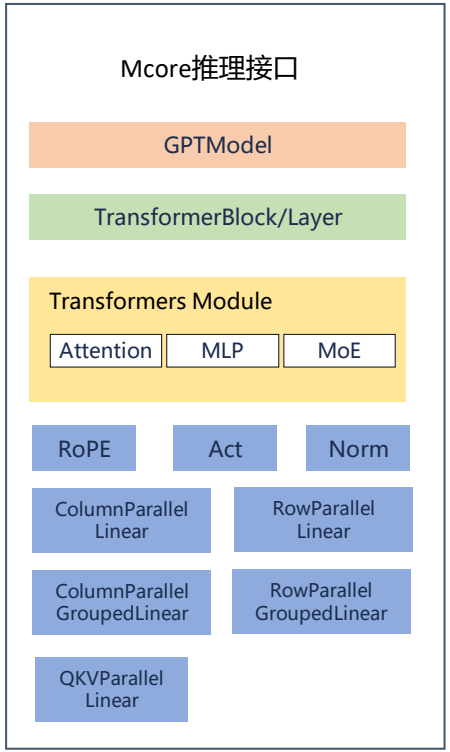
- 提供了预定义的主流 Transformer 模型架构。
- 主要子模块：
  - GPTModel: 用于实现类似 GPT 的自回归语言模型。
  - 模型层规范Spec构建: 用于实现大语言模型的泛化构建

Transformer 层 (parallel_core/inference /transformer)

- 这是整个库最核心的部分，它定义了 Transformer 架构的基本构建块，所有模型都是基于这个模块的组件构建的
- 提供模块化、可配置的 Transformer 组件。
- 主要子模块：
  - TransformerBlock/Layer： 代表一个完整的 Transformer 层
  - Attention/MLP/MOE： 实现多头注意力机制/前馈神经网络/专家模块，支持多种并行方式
  - LayerNorm/Activation/Rope：各种公共模块，归一化层、激活层、位置编码

分布式张量并行 (parallel_core/inference /tensor_parallel)

- 这是实现张量并行的核心，将模型层的权重在多张卡之间进行切分。
- 主要子模块：
  - ColumnParallelLinear/ColumnParallelGroupedLinear： 对线性层的权重矩阵按列切分。
  - RowParallelLinear/RowParallelGroupedLinear： 对线性层的权重矩阵按行切分。
  - QKVParallelLinear: 生成QKV矩阵，按照attention头进行切分



MindSpore Transformers Mcore推理模型库——Module架构

MLP

ColumnParallel Linear

RowParallel Linear

①定义数据类MLPSubmodules，表示MLP由哪些底层模块构成 （一个ColumnParallelLinear和RowParallelLinear）

②定义MLP类，实现__init__方法，传入子模块以及模块配置参数，定义模块结构

③实现MLP前向方法

```python
@dataclass
class MLPSubmodules:
    """
    The MLPSubmodules class defines two submodules for a Multi-Layer Perceptron (MLP):

    Args:
        linear_fc1 (Union[ModuleSpec, type], optional): The module definition for the first
            Defaults to None.
        linear_fc2 (Union[ModuleSpec, type], optional): The module definition for the second
            Defaults to None.
    """

    linear_fc1: Union[ModuleSpec, type] = None
    linear_fc2: Union[ModuleSpec, type] = None
```

```python
class MLP(nn.Cell):
    def __init__(
        self,
        config: TransformerConfig,
        submodules: MLPSubmodules,
        is_expert: bool = False,
        input_size: Optional[int] = None,
        delay_allreduce: bool = False,
        tp_group: Optional[ProcessGroup] = default_pgs,
        quant_config: Optional[QuantizationConfig] = None,
        prefix: str = "",
    ):
        super().__init__(config)

        self.config: TransformerConfig = config

        self.linear_fc1 = build_module(
            submodules.linear_fc1,
            self.input_size,
            ffn_hidden_size,
            config=self.config,
            bias=(self.config.add_bias_linear if self.config.add_mlp_fc1_bias_linear
                  is None else self.config.add_mlp_fc1_bias_linear),
            is_expert=is_expert,
            transpose_b=True,
            compute_dtype=self.config.compute_dtype,
            quant_config=quant_config,
            prefix=f"{prefix}.linear_fc1",
            **fc1_parallel_kwargs,
        )

        self.linear_fc2 = build_module(
            submodules.linear_fc2,
            self.config.ffn_hidden_size,
            self.config.hidden_size,
            config=self.config,
            bias=(self.config.add_bias_linear if self.config.add_mlp_fc2_bias_linear
                  is None else self.config.add_mlp_fc2_bias_linear),
            is_expert=is_expert,
            transpose_b=True,
            compute_dtype=self.config.compute_dtype,
            quant_config=quant_config,
            prefix=f"{prefix}.linear_fc2",
            **fc2_parallel_kwargs,
        )
```

MindSpore Transformers Mcore推理模型库——Module架构

SelfAttention

①定义数据类SelfAttentionSubmodules,表示SelfAttention由哪些底层模块构成

②定义SelfAttention类，实现__init__方法，传入子模块以及模块配置参数，定义模块结构

③实现前向方法

```python
@dataclass
class SelfAttentionSubmodules:
    """Configuration class for specifying the submodules of a self-attention."""

    linear_qkv: Union[ModuleSpec, type] = None
    core_attention: Union[ModuleSpec, type] = None
    linear_proj: Union[ModuleSpec, type] = None
    q_layernorm: Union[ModuleSpec, type] = None
    k_layernorm: Union[ModuleSpec, type] = None
```

```python
if self.use_flash_attention:
    self.core_attention = build_module(
        submodules.core_attention,
        head_num=self.num_attention_heads_per_partition,
        head_dim=self.hidden_size_per_attention_head,
        kv_head_num=self.num_query_groups_per_partition,
        scale_value=1.0 / self.norm_factor,
        next_tokens=0,
        pa_kv_head_num=self.num_query_groups_per_partition,
    )
```

```python
class SelfAttention(Attention):
    def __init__(self,
                 config: TransformerConfig,
                 submodules: SelfAttentionSubmodules,
                 layer_number: int,
                 attn_mask_type: AttnMaskType = None,
                 cp_comm_type: str = None,
                 delay_allreduce: bool = False,
                 model_comm_pgs: Optional[ModelCommProcessGroups] = default_model_comm_pgs,
                 quant_config: Optional[QuantizationConfig] = None,
                 prefix: str = ""):
        super().__init__(
            config=config,
            submodules=submodules,
            layer_number=layer_number,
            attn_mask_type=attn_mask_type,
            attention_type=None,
            cp_comm_type=cp_comm_type,
            delay_allreduce=delay_allreduce,
            model_comm_pgs=model_comm_pgs,
            quant_config=quant_config,
            prefix=prefix
        )

        self.linear_qkv = build_module(
            self.submodules.linear_qkv,
            self.config.hidden_size,
            self.hidden_size_per_attention_head,
            self.num_heads,
            self.config.num_query_groups,
            config=self.config,
            bias=self.config.add_qkv_bias,
            gather_output=False,
            transpose_b=True,
            compute_dtype=self.compute_dtype,
            tp_group=self.tp,
            quant_config=quant_config,
            prefix=f"{prefix}.linear_qkv",
        )
```

2. MindSpore Transformers Hugging Face模型迁移

MindSpore Transformers Hugging Face模型迁移——文件结构

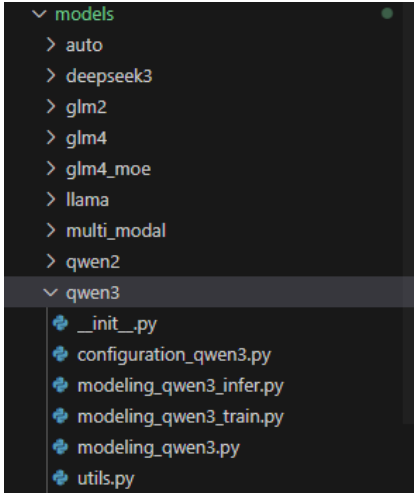
迁移总体流程

文件结构

模型代码放置于mindformers/models下，每个模型有独立文件夹

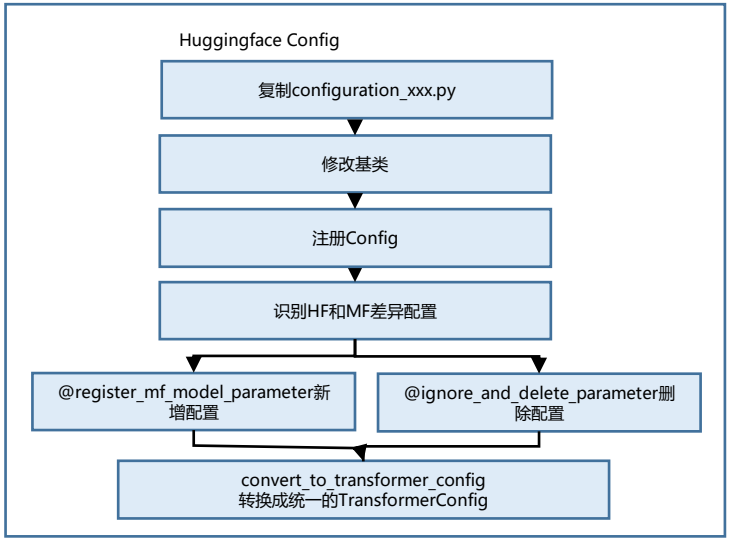
迁移HF模型时，主要需要创建以下文件：

- __init__.py
- configuration_xxx.py ---模型配置类
- modeling_xxx.py ---模型类
- modeling_xxx_infer.py ---推理模型类
- modeling_xxx_train.py ---训练模型类 （可选）
- utils.py



MindSpore Transformers Hugging Face模型迁移——HFConfig适配

1. 在configuration_xxx.py文件中，实现ModelConfig类。




```python
from mindformers.models.configuration_utils import PretrainedConfig
from mindformers.models.model_config_utils import (
    register_mf_model_parameter,
    ignore_and_delete_parameter,
    NotSupportedInfo
)
from mindformers.parallel_core.mf_model_config import MFModelConfig
from mindformers.tools.register import MindFormerRegister, MindFormerModuleType


zyw_hw, 3 months ago | 5 authors (sunyu-xuan and others)
@MindFormerRegister.register(MindFormerModuleType.CONFIG, legacy=False, search_names='qwen3')
class Qwen3Config(PretrainedConfig):
    """
    This is the configuration class to store the configuration of a [`Qwen3Model`]. It is used to instantiate a
    Qwen3 model according to the specified arguments, defining the model architecture. Instantiating a configuration
```

注册

修改基类

```python
@register_mf_model_parameter(
    mf_model_kwargs=MFModelConfig(
        pad_token_id=151643,
        block_size=32,
        num_blocks=1024,
        normalization='RMSNorm',
        add_bias_linear=False,
        gated_linear_unit=True,
        use_contiguous_weight_layout_attention=False
    ))
@ignore_and_delete_parameter(extra_ignore_param=[
    ('max_window_layers', NotSupportedInfo.useless),
    ('sliding_window', NotSupportedInfo.useless),
    ('use_sliding_window', NotSupportedInfo.useless),
    ('layer_types', NotSupportedInfo.useless),
])
def __init__(self,
             vocab_size=151936,
             hidden_size=4096,
             intermediate_size=22016,
             num_hidden_layers=32,
             num_attention_heads=32,
```

新增MF配置

删除配置

HuggingFace-Config	类型和默认值	MindFormers-Config（映射到TransformerConfig参数名称）
mlp_bias	bool: false	暂不支持
mlp_only_layers	list: []	暂不支持
sliding_window	null	暂不支持
output_router_logits	bool: false	暂不支持
router_aux_loss_coef	bool: true	暂不支持
decoder_sparse_step	int: 1	暂不支持
aux_loss_alpha	float: 0.001	暂不支持
seq_aux	bool: true	暂不支持
rope_scaling	"low_freq_factor": float,	不支持
	"high_freq_factor": float	不支持

HuggingFace-Config	类型和默认值	MindFormers-Config（映射到TransformerConfig参数名称）
use_cache	bool: true	use_past
torch_dtype	str: "bfloat16"	compute_dtype
max_window_layers	int	暂不支持
use_sliding_window	bool: false	暂不支持
quantization_config	dict:	暂不支持
	"activation_scheme": "dynamic"	暂不支持
	"fmt": "e4m3"	暂不支持
	"quant_method": "fp8"	暂不支持
	"weight_block_size"	暂不支持

MindSpore Transformers Hugging Face模型迁移——模型适配

2. 在utils.py文件中，实现模型基类。

①实现模型基类xxxPretrainedModel：继承PretrainedModel和ModelMixin类

② 指定config_class属性为模型对应Config类

```python
"""Qwen3 models' utils."""
from mindformers.models.qwen3.configuration_qwen3 import Qwen3Config
from mindformers.models.modeling_utils import PreTrainedModel
from mindformers.parallel_core.utils.model_mixin import ModelMixin
      zxq, 5 months ago • 【dev】【inference】qwen3_mcore模型适配

zxq, 3 months ago | 1 author (zxq)
class Qwen3PreTrainedModel(PreTrainedModel, ModelMixin):
    """
    An abstract class to handle weights initialization and a simple interface for downloading and loading pretrained
    models.
    """

    config_class = Qwen3Config
    base_model_prefix = "Qwen3"
```

继承基类

指定config类

MindSpore Transformers Hugging Face模型迁移——模型适配

2. 在modeling_xxx.py文件中，实现模型类。

① 实现最外层模型类：继承xxxPretrainedModel

② 注册模型

③ 实现__new__方法，根据不同的RUN_MODE，返回不同的模型

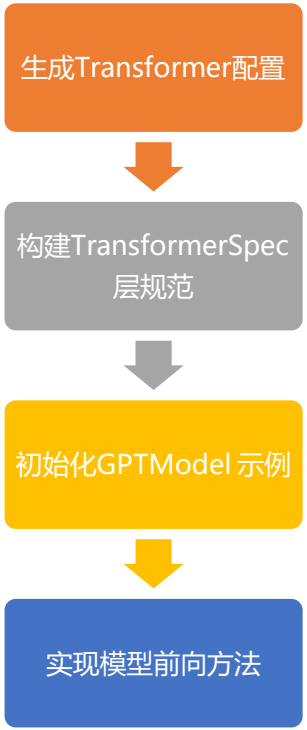
```
JavaZero, 4 months ago | 3 authors (zxq and others)
@MindFormerRegister.register(MindFormerModuleType.MODELS, legacy=False)
class Qwen3ForCausalLM(Qwen3PreTrainedModel):
    r"""
    Provide Qwen3 Model for training and inference.
    Args:
        config (Qwen3Config): The config of Qwen3 model.

    Returns:
        Tensor, the loss or logits of the network.
    """

    def __new__(cls, config):
        r"""
        get run mode to init different model.

        Args:
            config (Qwen3Config): The config of qwen3 model.

        Raises:
            NotImplementedError: Train mode is not supported for Qwen3 model.

        Returns:
            Tensor, the loss or logits of the network

        """
        if os.environ.get("RUN_MODE") == "predict":
            return InferenceQwen3ForCausalLM(config=config)
        return TrainingQwen3ForCausalLM(config=config)
```

注册模型

实现__new__方法

MindSpore Transformers Hugging Face模型迁移——模型适配

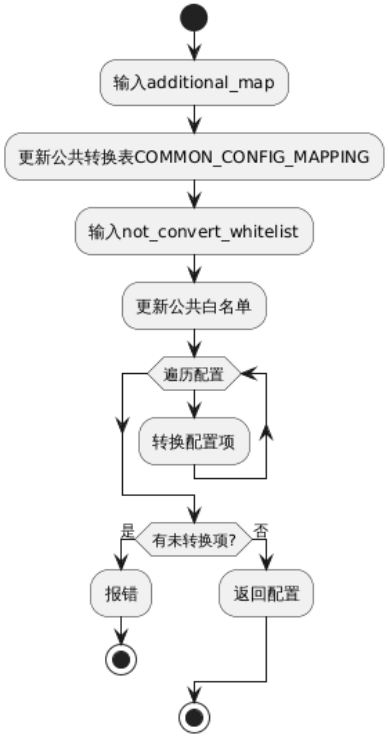
3、在modeling_xxx_infer.py文件中，实现推理模型类




```python
class InferenceQwen3ForCausalLM(Qwen3PreTrainedModel, InferModelMixin):

    def __init__(self, config: Qwen3Config):
        super().__init__(config, auto_prefix=False)
        self.config = config
        # step1: 由模型config转换生成transformer config
        config: TransformerConfig = self.convert_to_transformer_config(self.config)
        # 可选 量化推理配置
        self.quant_config = get_quant_config(self.config)
        # step2: 构建TransformerSpec层规范
        transformer_layer_spec=get_gpt_layer_local_spec(
                                normalization=config.normalization,
                                use_flash_attention=self.config.use_flash_attention,
                                qk_layernorm=True,
                            )
        # step3: 初始化GPTModel实例
        self.model = GPTModel(
            config=config,
            transformer_layer_spec=transformer_layer_spec,
            vocab_size=self.vocab_size,
            max_sequence_length=self.max_position_embeddings,
            position_embedding_type=config.position_embedding_type,
            rotary_base=self.config.rope_theta,
            share_embeddings_and_output_weights=self.config.tie_word_embeddings,
            pre_process=config.pre_process,
            post_process=config.post_process,
            quant_config=self.quant_config
        )
    # step4: 实现模型前向方法
    def construct(self, input_ids, hidden_states=None):
        logits = self.model(input_ids=input_ids,hidden_states=hidden_states)
        return logits
```

MindSpore Transformers Hugging Face模型迁移——模型适配

convert_to_transformer_config()的工作流



公共转换表 COMMON_CONFIG_MAPPING

位置： mindformers\parallel_core\transformer_config_utils.py

```
# Model Architecture
("mtp_depth", "num_nextn_predict_layers", "mtp_num_layers"): "mtp_num_layers",
("mtp_loss_factor", "mtp_loss_scaling_factor"): "mtp_loss_scaling_factor",
("num_heads", "num_attention_heads", "n_head"): "num_attention_heads",
("n_kv_heads", "num_key_value_heads", "num_query_groups"): "num_query_groups",
("intermediate_size", "ffn_hidden_size"): "ffn_hidden_size",
("head_dim", "kv_channels"): "kv_channels",
```

使用示例

```
"""
假设XXXConfig存在特殊配置。

覆盖公共转换：mtp_layers, compute_in_fp32

无用配置：kv_cache, top_k
"""

def is_fp32(compute_in_fp32):
  if compute_in_fp32:
    return float32
  return bfloat16

class TrainingXXXForCausalLM(TrainModelMixin, XXXPreTrainedModel):
    def __init__(self, config: XXXConfig):
        super().__init__(config, auto_prefix=False)

        transformer_config = self.convert_to_transformer_config(config,
                                                                 is_mla_model=True,
                                                                 additional_map={"mtp_layers": "mtp_num_layers",
                                                                                 "compute_in_fp32": ("compute_dtype", is_fp32)
                                                                                 },
                                                                 not_convert_whitelist={"kv_cache", "top_k"}
                                                                )
```

MindSpore Transformers Hugging Face模型迁移——模型适配

- Transformer大模型在模型结构上具有相似性，MCore通过以下两个核心函数实现大语言模型的泛化构建：

✓ get_gpt_layer_local_spec: 构建单个 Transformer 层的模块规范，适用于每层结构相同的模型搭建（如Qwen3）

✓ get_gpt_decoder_block_spec: 构建整个解码器块的模块规范，即多层Transformer模块，支持MoE、Dense混合层构建，适用于存在不同Transformer层结构的模型搭建（如DeepSeekV3）

- ModuleSpec是模型组装的实现类，可以通过指定module和submodules参数来动态创建我们想要的层

get_gpt_layer_local_spec

该函数用于定义单个Transformer层的模块规范，支持多种结构特性：

- MLP（Multilayer Perceptron）
- MoE（Mixture of Experts）
- GQA/MHA
- Multi Latent Attention
- Q/K LayerNorm
- Sandwich Norm

```
self_attn = ModuleSpec(
    module=SelfAttention,
    submodules=SelfAttentionSubmodules(
        core_attention=FlashAttention if use_flash_attention else DotProductAttention,
        linear_proj=RowParallelLinear,
        linear_qkv=QKVParallelLinear,
        q_layernorm=get_norm_cls(normalization) if qk_layernorm else IdentityOp,
        k_layernorm=get_norm_cls(normalization) if qk_layernorm else IdentityOp,
    )
)

mlp = ModuleSpec(
    module=MLP,
    submodules=MLPSubmodules(
        linear_fc1=MergedColumnParallelLinear if gated_linear_unit else ColumnParallelLinear,
        linear_fc2=RowParallelLinear,
    ),
)

transformer_layer_spec = ModuleSpec(
    module=TransformerLayer,
    submodules=TransformerLayerSubmodules(
        input_layernorm=get_norm_cls(normalization),
        self_attention=self_attention,
        post_self_attn_layernorm=get_norm_cls(normalization) if sandwich_norm else IdentityOp,
        pre_mlp_layernorm=get_norm_cls(normalization),
        mlp=mlp,
        post_mlp_layernorm=get_norm_cls(normalization) if sandwich_norm else IdentityOp,
    )
)
```

MindSpore Transformers Hugging Face模型迁移——模型适配

- Transformer大模型在模型结构上具有相似性，MCore通过以下两个核心函数实现大语言模型的泛化构建：

✓ get_gpt_layer_local_spec: 构建单个 Transformer 层的模块规范，适用于每层结构相同的模型搭建（如Qwen3）

✓ get_gpt_decoder_block_spec: 构建整个解码器块的模块规范，即多层Transformer模块，支持MoE、Dense混合层构建，适用于存在不同Transformer层结构的模型搭建（如DeepSeekV3）

get_gpt_decoder_block_spec

该函数用于构建整个 Transformer解码器块结构，支持：

- MoE和Dense层混合搭建
- 读取配置信息定义层间排布

```python
def get_gpt_decoder_block_spec(
        config: TransformerConfig,
        normalization: Optional[str] = None,
        qk_l2_norm: Optional[bool] = False,
) -> TransformerLayerSubmodules:
    """GPT block spec."""
    # layer specs.
    dense_layer_spec = get_gpt_layer_local_spec(
        num_experts=None,
        moe_grouped_gemm=False,
        qk_layernorm=config.qk_layernorm,
        gated_linear_unit=config.gated_linear_unit,
        multi_latent_attention=config.multi_latent_attention,
        normalization=normalization,
        use_flash_attention=config.use_flash_attention,
        qk_l2_norm=qk_l2_norm,
        use_alltoall=config.use_alltoall,
        use_fused_mla=config.use_fused_mla,
    )

    moe_layer_spec = get_gpt_layer_local_spec(
        num_experts=config.num_moe_experts,
        moe_grouped_gemm=True,
        qk_layernorm=config.qk_layernorm,
        gated_linear_unit=config.gated_linear_unit,
        multi_latent_attention=config.multi_latent_attention,
        normalization=normalization,
        use_flash_attention=config.use_flash_attention,
        qk_l2_norm=qk_l2_norm,
        use_alltoall=config.use_alltoall,
        use_fused_mla=config.use_fused_mla,
    )
```

MindSpore Transformers Hugging Face模型迁移——模型适配

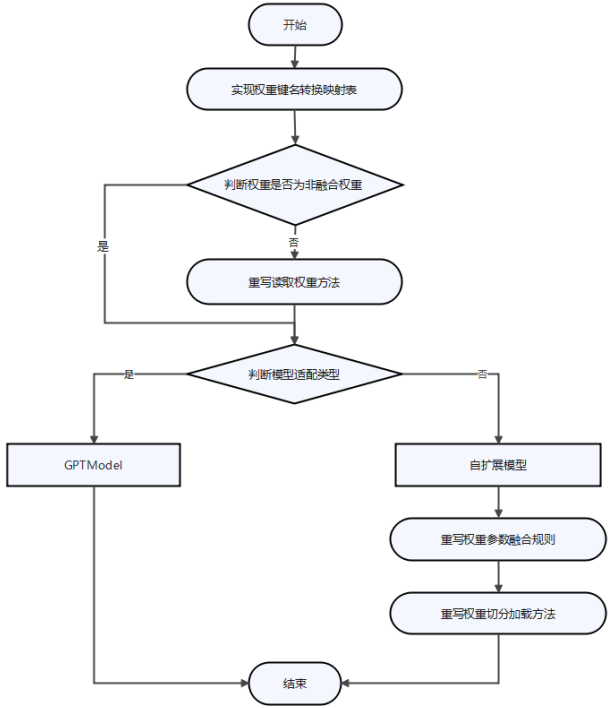
GPTModel作为MCore大语言模型统一入口，支持Transformer结构的模型构建能力，其init入参和调用示例如下

参数名	类型	描述	默认值
config	TransformerConfig	模型全局配置对象，包含隐藏层大小、注意力头数等核心参数	-
transformer_layer_spec	ModuleSpec	定义Transformer层结构的模块规范，指定SelfAttention/MLP等组件	-
vocab_size	int	词汇表大小，决定嵌入层和输出层的维度	-
max_sequence_length	int	序列最大长度，用于位置编码的初始化	-
pre_process	bool	是否执行嵌入层计算（当前仅支持True）	True
post_process	bool	是否计算输出层logits（默认启用）	True
parallel_output	bool	是否分散输出（保持张量并行分片）	True
share_embeddings_and_output_weights	bool	是否共享嵌入层与输出层权重（当前不支持）	False
position_embedding_type	str	位置编码类型	'learned_absolute'
rotary_percent	float	应用旋转编码的维度比例	1.0
rotary_base	int	RoPE基础值	10000
rope_scaling	bool	是否启用RoPE外推算法（通过rposition_embedding_type='llama3'控制）	False
seq_len_interpolation_factor	float	序列长度插值因子	None

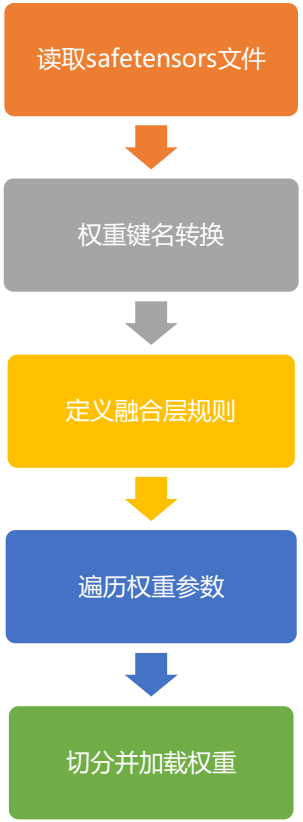
3. MindSpore Transformers Hugging Face权重加载

MindSpore Transformers 模型权重加载全流程

- 支持直接加载huggingface权重
- 可以处理多种泛化场景权重



MindSpore Transformers 模型权重加载全流程




```python
with safe_open(st_file, framework="np") as f:
# step1: 读取safetensors文件
    for name in f.keys():
        param = f.get_slice(name)
        # step2: 权重键名转换
        name = self.convert_name(name)
        yield name, param

# step3: 定义融合层规则
# 非MoE路由专家但需融合的权重键名部分映射规则:
[
    (".linear_qkv", ".linear_q", "q"),
    (".linear_qkv", ".linear_k", "k"),
    (".linear_qkv", ".linear_v", "v"),
    (".mlp.linear_fc1", ".mlp.gating", "gating"),
    (".mlp.linear_fc1", ".mlp.linear_fc1", "hidden"),
]

# MoE中与路由专家相关的权重键名部分映射规则:
[
    ("experts.weight1", "experts.{expert_id}.gating.", "{expert_id}","w1"),
    ("experts.weight1", "experts.{expert_id}.linear_fc1.", "{expert_id}","w3"),
    ("experts.weight2", "experts.{expert_id}.linear_fc2.", "{expert_id}","w2"),
]

# step4: 遍历权重参数
for name, loaded_weight in weights:
    if name in params_dict:
        param = params_dict[name]
        weight_loader = param.weight_loader
        # step5: 切分并加载权重
        weight_loader(param, loaded_weight, shard_id)
```

MindSpore Transformers Hugging Face模型权重加载

1. 在utils.py文件中，实现权重键名映射表。

元组左边为Hugging Face权重键名，右边为网络中的键名

```python
class Qwen3PreTrainedModel(PreTrainedModel, ModelMixin):
weight_mapping = [
        ('model.embed_tokens.', 'embedding.word_embeddings.'),
        ('.self_attn.q_proj.', '.self_attention.linear_q.'),
        ('.self_attn.k_proj.', '.self_attention.linear_k.'),
        ('.self_attn.v_proj.', '.self_attention.linear_v.'),
        ('.mlp.gate_proj.', '.mlp.gating.'),
        ('.mlp.down_proj.', '.mlp.linear_fc2.'),
        ('.mlp.up_proj.', '.mlp.hidden.'),
]
```

```python
# 获取网络中所有模块的键名
def get_params_dict(self):
    params_dict = {}
    for _, module in self.modules_dict.items():
        module_params = module.parameters_dict()
        for param_name, param in module_params.items():
            params_dict[param_name] = param

    return params_dict
```

MindSpore Transformers Hugging Face模型权重加载

- 提供统一接口_safetensors_weights_iterator 实现权重前处理
- 复用同一套权重加载流程，减少适配工作

2. 在modeling_xxx.py.py文件中，重写读取权重方法

将Hugging Face telechat2中已融合的权重进行拆分

```python
class InferenceTelechat2ForCausalLM(Telechat2PreTrainedModel, InferModelMixin):
    def _safetensors_weights_iterator(self, weights_files: List[str]) -> Generator[Tuple[str, Any], None, None]:
        """Iterate over the weights in the model safetensor files."""
        rank_id = get_tensor_model_parallel_rank()
        is_main_rank = rank_id == 0
        for st_file in tqdm(
                weights_files,
                desc=f"[Rank {rank_id}] Loading safetensors checkpoint shards",
                disable=not is_main_rank
        ):
            with safe_open(st_file, framework="np") as f:
                for name in f.keys():  # noqa: SIM118
                    # Return a lightweight PySafeSlice object
                    # that uses file pointer offset internally to read Safetensor
                    # on demand, avoiding memory explosion. Actual data can be obtained through slicing operation
                    # like param[start:end]
                    param = f.get_slice(name)
                    name = self.convert_name(name)
                    if ".key_value." in name and self.quant_config is None and self.config.quantization is None:
                        num_heads = self.config.n_head
                        n_kv_heads = self.config.num_key_value_heads
                        hidden_size = self.config.hidden_size
                        head_dim = hidden_size // num_heads
                        n_rep = num_heads // n_kv_heads
                        kv_channel = hidden_size // n_rep
                        param = np.array(param[:])
                        param = param.reshape((n_kv_heads, 2*head_dim, -1))
                        k_key = name.replace("key_value", "linear_k")
                        k_weight = param[:, :head_dim, :].reshape((kv_channel, -1))
                        v_key = name.replace("key_value", "linear_v")
                        v_weight = param[:, head_dim:2*head_dim, :].reshape((kv_channel, -1))
                        yield k_key, k_weight
                        yield v_key, v_weight
                    else:
                        yield name, param
```

MindSpore Transformers Hugging Face模型权重加载 --扩展

3. 重写参数融合规则

```
# 非MoE路由专家但需融合的权重键名部分映射规则:
[
    (".linear_qkv", ".linear_q", "q"),
    (".linear_qkv", ".linear_k", "k"),
    (".linear_qkv", ".linear_v", "v"),
    (".mlp.linear_fc1", ".mlp.gating", "gating"),
    (".mlp.linear_fc1", ".mlp.linear_fc1", "hidden"),
]
```

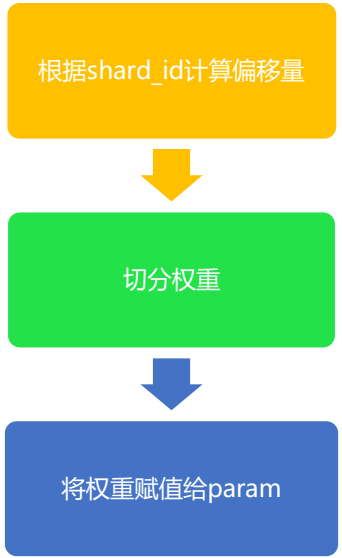
每一个元组中：
第一个元素是网络中融合层的权重名
第二个元素是Hugging Face权重转换后的权重键名
第三个元素是标志位，表示当前加载的权重是对应qkv中的哪一个，根据这个标志位进行后续权重切分与加载

```
# MoE中与路由专家相关的权重键名部分映射规则:
[
    ("experts.weight1", "experts.{expert_id}.gating.", "{expert_id}","w1"),
    ("experts.weight1", "experts.{expert_id}.linear_fc1.", "{expert_id}","w3"),
    ("experts.weight2", "experts.{expert_id}.linear_fc2.", "{expert_id}","w2"),
]
```

每一个元组中：
第一个元素是网络中融合层的权重名
第二个元素是Hugging Face权重转换后的权重键名
第三个元素是专家id，标识第几个专家
第四个元素是标志位，根据这个标志位进行后续权重切分与加载

MindSpore Transformers Hugging Face模型权重加载 --扩展

4. 重写模块的权重切分加载方法




```python
def weight_loader(self, param, loaded_weight, loaded_shard_id: Optional[str] = None):
        output_dim = getattr(param, "output_dim", None)
        tp_rank = self.tp_group.rank
        if loaded_shard_id is not None:
            if loaded_shard_id == "q":
                shard_offset = 0
                shard_size = self.num_heads * self.head_size
            elif loaded_shard_id == "k":
                shard_offset = self.num_heads * self.head_size
                shard_size = self.num_kv_heads * self.head_size
            elif loaded_shard_id == "v":
                shard_offset = (self.num_heads +self.num_kv_heads) * self.head_size
                shard_size = self.num_kv_heads * self.head_size
            shard_id = tp_rank
            start_idx = shard_id * shard_size
            # 权重在并行tp域切分
            loaded_weight = split_loaded_weight(loaded_weight, output_dim, start_idx, shard_size)
            param.asnumpy()[shard_offset:shard_offset + shard_size] = loaded_weight
```

4. MindSpore Transformers 模型迁移案例

Qwen系列模型迁移案例

Qwen2、Qwen3、Qwen3Moe目前已上线MindSpore Transformer，可参考对应modeling_infer.py代码实现

https://gitee.com/mindspore/mindformers/tree/master/mindformers/models/qwen2

https://gitee.com/mindspore/mindformers/tree/master/mindformers/models/qwen3

https://gitee.com/mindspore/mindformers/tree/master/mindformers/models/qwen3_moe

Qwen2

- 自注意力模块（multi_latent_attention=False）
- 前馈网络MLP模块（num_experts=None）
- RMSNorm模块（normalization="RMSNorm"）

Qwen3

- 自注意力模块（multi_latent_attention=False）
- 前馈网络MLP模块（num_experts=None）
- RMSNorm模块（normalization="RMSNorm"）
- qk norm结构（qk_layernorm=True）

Qwen3 Moe

- 自注意力模块（multi_latent_attention=False）
- MoE模块（num_experts=128）
- RMSNorm模块（normalization="RMSNorm"）
- qk norm结构（qk_layernorm=True）

```python
class InferenceQwen2ForCausalLM(Qwen2PreTrainedModel, InferModelMixin):
    r"""
    Provide qwen2 model infer through network.

    Args:
        config (Qwen2Config): The config of qwen2 model.

    Returns:
        output: Tensor, the output of qwen2 decoder layer

    """

    def __init__(self, config: Qwen2Config):
        super().__init__(config, auto_prefix=False)
        self.config = config
        config: TransformerConfig = self.convert_to_transformer_config(self.config)

        self.pad_token_id = self.config.pad_token_id
        self.vocab_size = config.vocab_size
        self.max_position_embeddings = config.max_position_embeddings
        self.compute_dtype = config.compute_dtype

        self.is_prefill = True
        self.model = GPTModel(config=config,
                              transformer_layer_spec=get_gpt_layer_local_spec(
                                  normalization=config.normalization,
                                  use_flash_attention=self.config.use_flash_attention,
                                  qk_layernorm=False,
                              ),
                              vocab_size=self.vocab_size,
                              max_sequence_length=self.max_position_embeddings,
                              position_embedding_type=config.position_embedding_type,
                              rotary_base=self.config.rope_theta,
                              share_embeddings_and_output_weights=self.config.tie_word_embeddings,
                              pre_process=self.config.pre_process,
                              post_process=self.config.post_process,)
```

MindSpore Transformers大模型使能教程：大模型Mcore内测教程

Qwen系列模型迁移按例

Qwen2、Qwen3、Qwen3Moe目前已上线MindSpore Transformer，可参考对应modeling_infer.py代码实现

https://gitee.com/mindspore/mindformers/tree/master/mindformers/models/qwen2

https://gitee.com/mindspore/mindformers/tree/master/mindformers/models/qwen3

https://gitee.com/mindspore/mindformers/tree/master/mindformers/models/qwen3_moe

Qwen2

- 自注意力模块（multi_latent_attention=False）
- 前馈网络MLP模块（num_experts=None）
- RMSNorm模块（normalization="RMSNorm"）

Qwen3

- 自注意力模块（multi_latent_attention=False）
- 前馈网络MLP模块（num_experts=None）
- RMSNorm模块（normalization="RMSNorm"）
- qk norm结构（qk_layernorm=True）

Qwen3 Moe

- 自注意力模块（multi_latent_attention=False）
- MoE模块（num_experts=128）
- RMSNorm模块（normalization="RMSNorm"）
- qk norm结构（qk_layernorm=True）

```python
class InferenceQwen3ForCausalLM(Qwen3PreTrainedModel, InferModelMixin):
    r"""
    Provide qwen3 model infer through network.

    Args:
        config (Qwen3Config): The config of qwen3 model.

    Returns:
        output: Tensor, the output of qwen3 decoder layer

    """

    def __init__(self, config: Qwen3Config):
        super().__init__(config, auto_prefix=False)
        self.config = config
        config: TransformerConfig = self.convert_to_transformer_config(self.config)

        self.quant_config = get_quant_config(self.config, self.weight_mapping)
        self.pad_token_id = self.config.pad_token_id
        self.vocab_size = config.vocab_size
        self.max_position_embeddings = config.max_position_embeddings
        self.compute_dtype = config.compute_dtype

        self.is_prefill = True
        self.model = GPTModel(config=config,
                              transformer_layer_spec=get_gpt_layer_local_spec(
                                  normalization=config.normalization,
                                  use_flash_attention=self.config.use_flash_attention,
                                  qk_layernorm=True,
                              ),
                              vocab_size=self.vocab_size,
                              max_sequence_length=self.max_position_embeddings,
                              position_embedding_type=config.position_embedding_type,
                              rotary_base=self.config.rope_theta,
                              share_embeddings_and_output_weights=self.config.tie_word_embeddings,
                              pre_process=config.pre_process,
                              post_process=config.post_process,
                              quant_config=self.quant_config)
```

MindSpore Transformers大模型使能教程：大模型Mcore内测教程

Qwen系列模型迁移实践

Qwen2、Qwen3、Qwen3Moe目前已上线MindSpore Transformer，可参考对应modeling_infer.py代码实现

https://gitee.com/mindspore/mindformers/tree/master/mindformers/models/qwen2

https://gitee.com/mindspore/mindformers/tree/master/mindformers/models/qwen3

https://gitee.com/mindspore/mindformers/tree/master/mindformers/models/qwen3_moe

Qwen2

- 自注意力模块（multi_latent_attention=False）
- 前馈网络MLP模块（num_experts=None）
- RMSNorm模块（normalization="RMSNorm"）

Qwen3

- 自注意力模块（multi_latent_attention=False）
- 前馈网络MLP模块（num_experts=None）
- RMSNorm模块（normalization="RMSNorm"）
- qk norm结构（qk_layernorm=True）

Qwen3 Moe

- 自注意力模块（multi_latent_attention=False）
- MoE模块（num_experts=128）
- RMSNorm模块（normalization="RMSNorm"）
- qk norm结构（qk_layernorm=True）

```python
class InferenceQwen3MoeForCausalLM(Qwen3MoePreTrainedModel, InferModelMixin):
    r"""
    Provide qwen3_moe model infer through network.

    Args:
        config (Qwen3MoeConfig): The config of qwen3_moe model.

    Returns:
        output: Tensor, the output of qwen3_moe decoder layer

    """

    def __init__(self, config: Qwen3MoeConfig):
        super().__init__(config, auto_prefix=False)
        self.config = config
        config: TransformerConfig = self.convert_to_transformer_config(self.config)

        # update communication-related configuration in TransformerConfig
        config = update_comm_config(config)
        self.pad_token_id = self.config.pad_token_id
        self.vocab_size = config.vocab_size
        self.max_position_embeddings = config.max_position_embeddings
        self.compute_dtype = config.compute_dtype
        self.is_prefill = True
        self.model = GPTModel(config=config,
                              transformer_layer_spec=get_gpt_layer_local_spec(
                                  num_experts=config.num_moe_experts,
                                  normalization=config.normalization,
                                  use_flash_attention=self.config.use_flash_attention,
                                  qk_layernorm=True,
                                  use_alltoall=config.use_alltoall,
                              ),
                              vocab_size=self.vocab_size,
                              max_sequence_length=self.max_position_embeddings,
                              position_embedding_type=config.position_embedding_type,
                              rotary_base=self.config.rope_theta,
                              share_embeddings_and_output_weights=self.config.tie_word_embeddings,
                              pre_process=config.pre_process,
                              post_process=config.post_process,)
```

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谢谢观看！

MindSpore Transformers LLM推理迁移与实践

MindSpore Transformers 大模型系列课程